

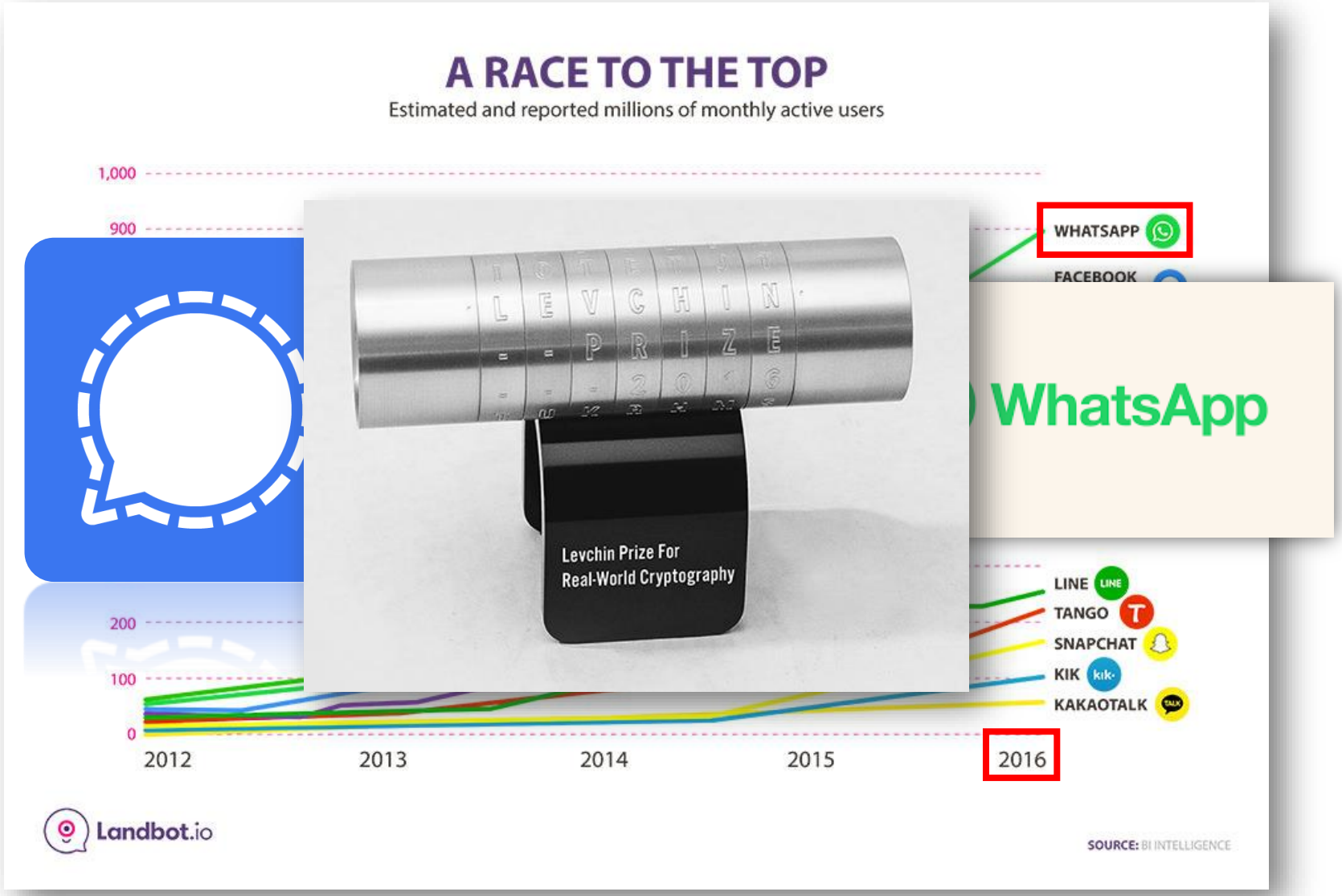
ΑΡΙΣΤΟΤΕΛΕΙΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΘΕΣΣΑΛΟΝΙΚΗΣ
ΣΧΟΛΗ ΘΕΤΙΚΩΝ ΕΠΙΣΤΗΜΩΝ
ΤΜΗΜΑ ΜΑΘΗΜΑΤΙΚΩΝ



An overview of the Signal Protocol

ΦΛΩΡΙΑΣ ΠΑΠΑΔΟΠΟΥΛΟΣ

Importance of Signal Protocol



PRELIMINARIES

Curve Models

Definition 1.2. A Weierstrass equation of an elliptic curve E over a field K is

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6,$$

where $a_1, a_2, a_3, a_4, a_6 \in K$ and $\Delta \neq 0$ where Δ denotes the discriminant of E.

Other models ?

Edwards Montgomery

- Birational Equivalence:

$$E_1 \underset{\varphi_2}{\overset{\varphi_1}{\rightleftarrows}} E_2$$

Edwards Curves

$char(K) \neq 2$

Edwards Curves

$$x^2 + y^2 = 1 + dx^2y^2$$

$d \neq \{0,1\}$

Twisted Edwards Curves

$$ax^2 + y^2 = 1 + dx^2y^2$$

$a \neq d \ \& \ d \neq 0$

K-Complete twisted Edwards Curves

$$ax^2 + y^2 = 1 + dx^2y^2$$

a: square & d: nonsquare

Twisted Edwards Curves

$$ax^2 + y^2 = 1 + dx^2y^2$$

a: square & d: nonsquare

- (Complete) Addition Law:

$$O = (0,1)$$

$$-(x, y) = (-x, y)$$

$$(x_1, y_1) + (x_2, y_2) = \left(\frac{x_1y_1 + x_2y_1}{1 + dx_1x_2y_1y_2}, \frac{y_1y_2 - ax_1x_2}{1 - dx_1x_2y_1y_2} \right)$$

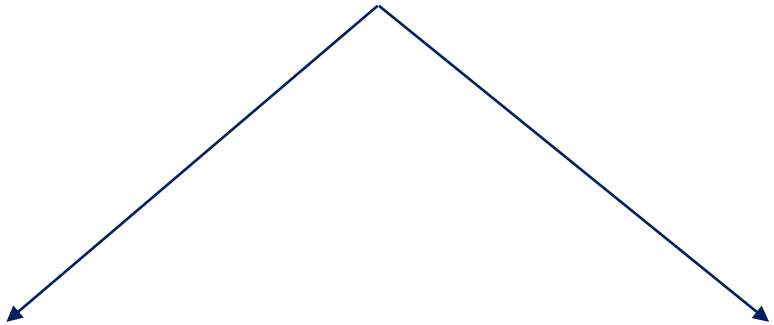
Montgomery Curves

Montgomery Curves

$$Bv^2 = u^3 + Au^2 + u$$

$$\mathrm{B} \neq 0 \ \& \ \mathrm{A} \neq \pm 2$$

$char(K) \neq 2$



Montgomery Ladder

$\subseteq$ Weierstrass curves

$\Rightarrow$ Same Addition Law


Twisted Edwards
Montgomery
EdDSA
ECDH

Safe Curves

Curve25519

128-bit security

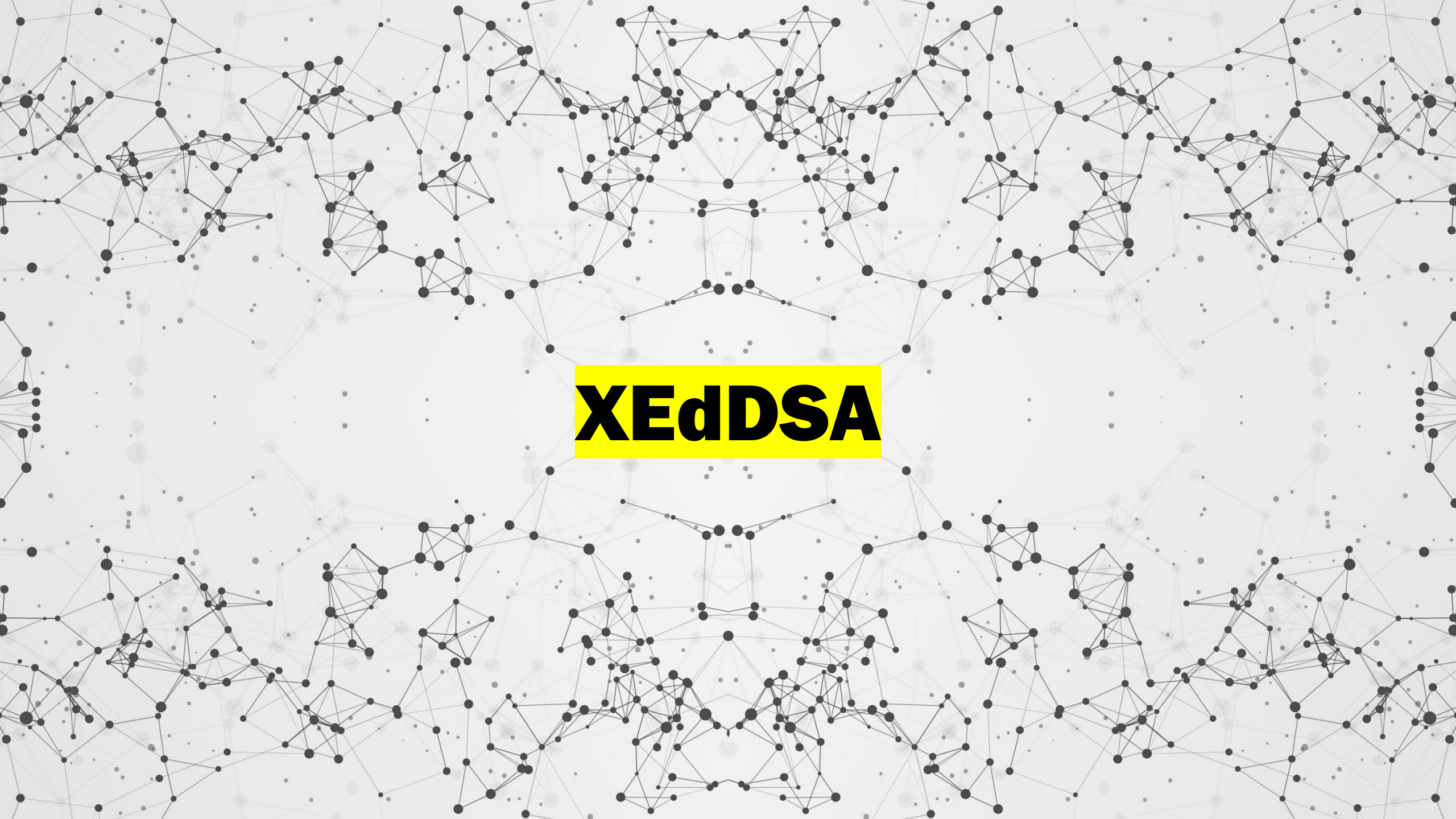
X25519

Curve448

224-bit security

X448

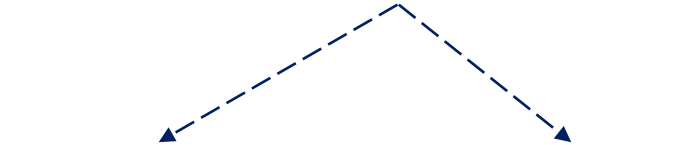
"The 224-bit security level of Curve448 strikes a delicate balance between performance and paranoia."


XEdDSA

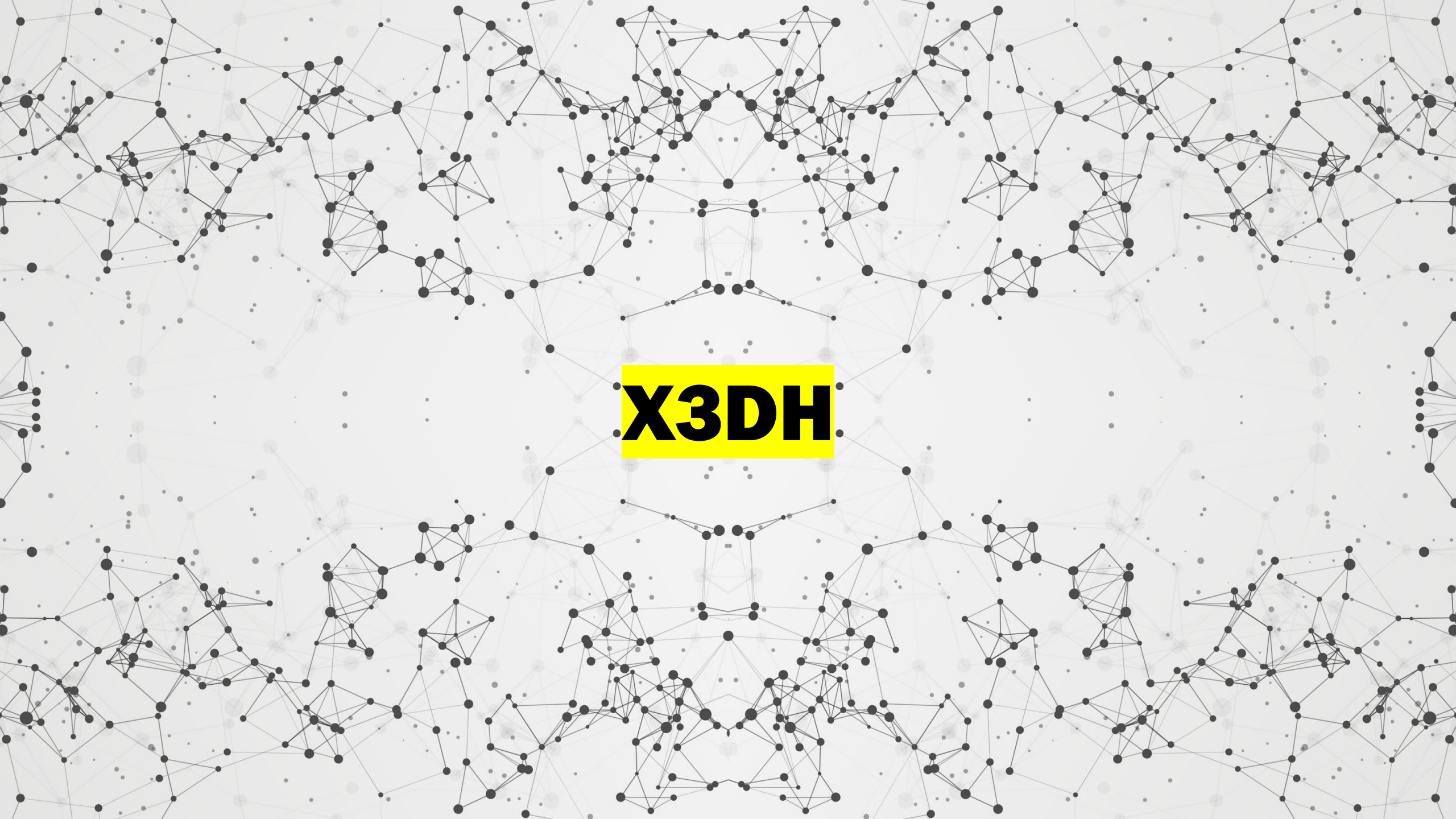
XEdDSA

EdDSA = ECDSA + Twisted Edwards Curves

XEdDSA = EdDSA + Montgomery keys for ECDH

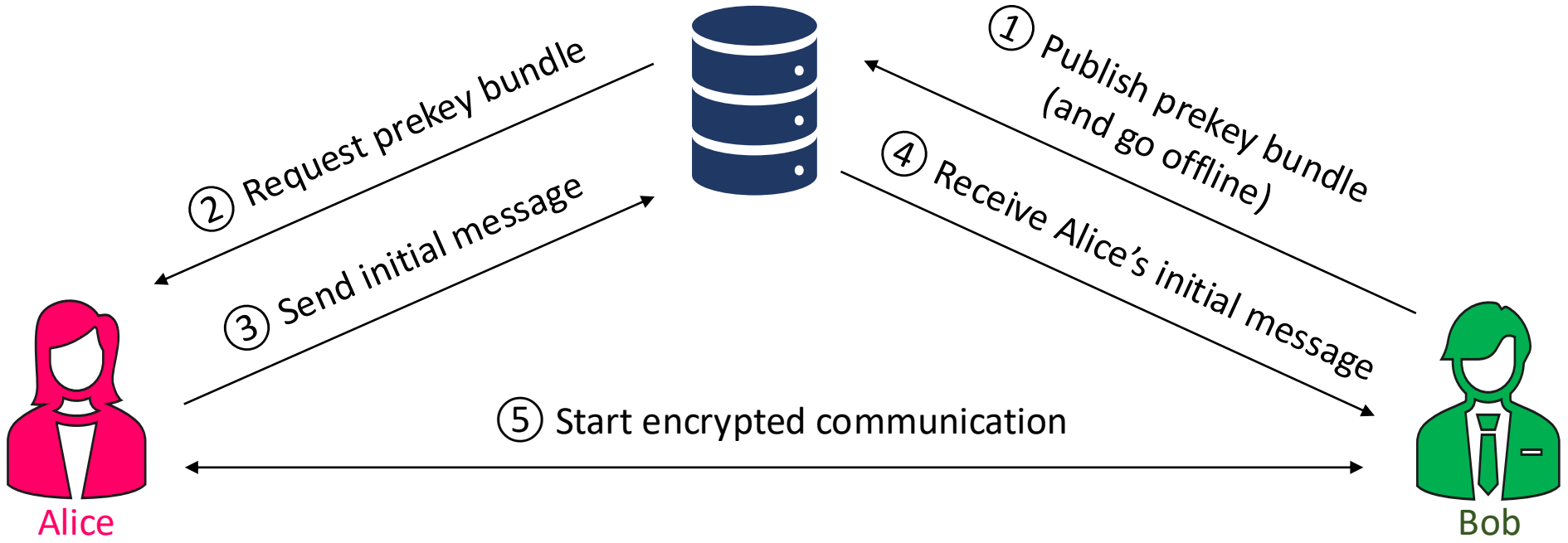


XEd25519 XEd448


X3DH

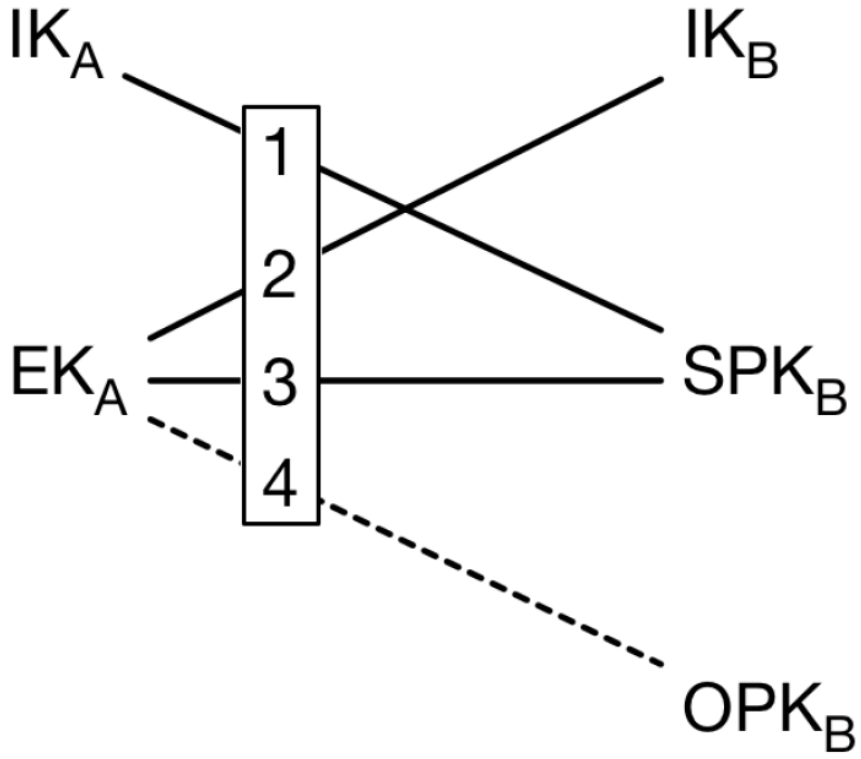
X3DH

- X3DH: Asynchronous Key Agreement Protocol

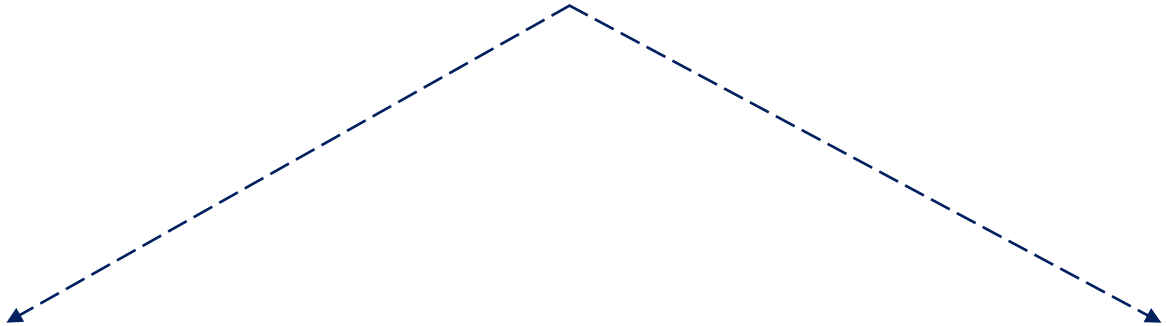


X3DH

Public key	Definition
IK_A	Alice's identity key
EK_A	Alice's ephemeral key
IK_B	Bob's identity key
SPK_B	Bob's signed prekey
OPK_B	Bob's one-time prekey

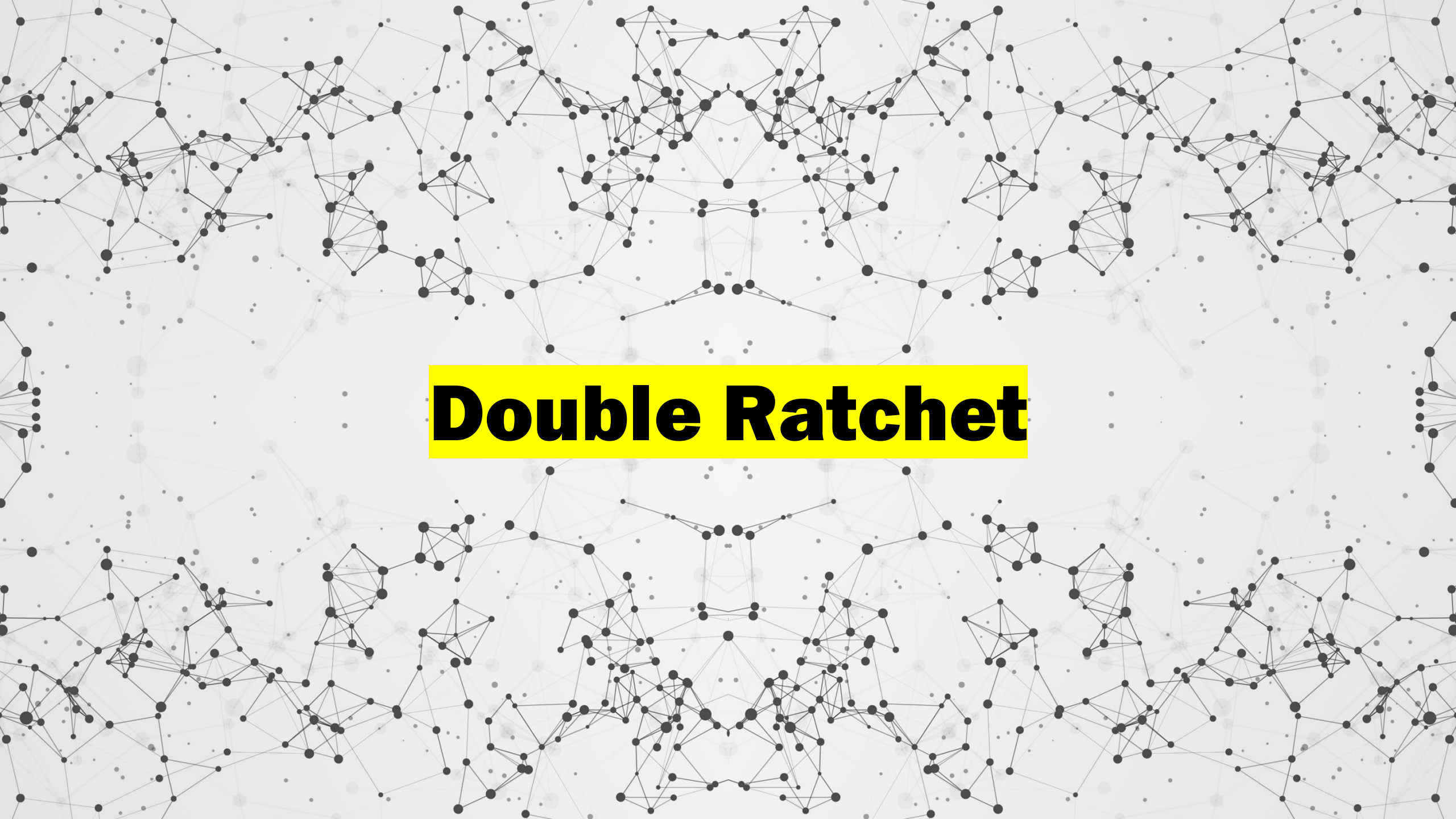


Why X3DH?



Forward Secrecy

Cryptographic Deniability


Double Ratchet

ΕΥΧΑΡΙΣΤΩ